

DiaTwin - System architecture

Type 2 Diabetes digital twin: fusion of static EHR and dynamic wearable/CGM streams

1 DATA STREAMS

Static / historical

Synthea-style simulated EHR
demographics, HbA1c, eGFR, LDL
TCF7L2 marker, medication
comorbidities, lifestyle

Dynamic / real-time

CGM glucose (5 min, with gaps)
heart rate, HRV (RMSSD)
steps, sleep stages
meal diary (logged carbs)

2 FUSION

Fusion layer

align streams per patient
glucose lags, trends, stats
sleep / HRV / activity
meal timing, time of day
+ static EHR features
= 43-feature vector
missing streams stay NaN

3 TWIN CORE

Learned layer

Gradient-boosted trees
30 / 60 / 120 min forecasts
10-90% band, spike + hypo risk
modality-dropout training

DIGITAL TWIN (per patient)

Mechanistic layer

Personal insulin sensitivity
(derived from the EHR)
meal + exercise kernels
what-if simulation

4 CLINICIAN OUTPUT

Alerts

hyper / hypo, LOW-WATCH-ALERT

Explanations

why: feature-group effects

What-if planner

eat / half / walk / skip

Cohort + data quality

risk board, stream health

Streamlit dashboard

doctor-facing virtual-patient app

Rigorous evaluation

patient-level split, CIs, subgroups, robustness

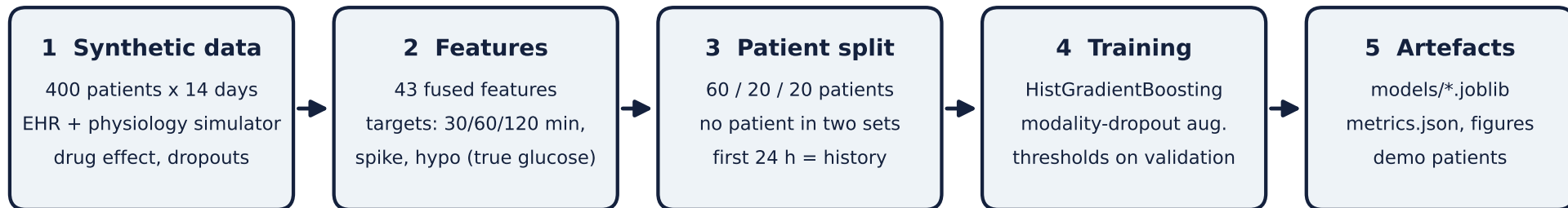
Privacy by design

synthetic data only (DPDP Act / HIPAA safe)

DiaTwin - ML pipeline and evaluation

Reproducible with one command: `python -m src.train`

BUILD



EVALUATION SUITE (all patient-level)



Serving: Streamlit dashboard loads the saved models and replays each virtual patient's stream. CI: GitHub Actions runs the test-suite on every push.